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In-Field Evaluation of a Detector-Free Vision Pipeline for Cabbage Crop-Row Guidance

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Abstract: Vision-only crop-row guidance lets a low-cost field robot steer without RTK-GPS, but for every frame it must produce the heading and cross-track offset that drive the steering controller. Most field systems obtain these quantities from a trained row detector, which needs field-specific labelling and tends to degrade when the canopy or growth stage drifts away from the training distribution. We ask whether a detector-free and training-free pipeline built from classical components can meet the same interface, and we evaluate one in the field on a cabbage robot. A forward-view camera feeds an Excess-Green central-row tracker, and a robust quadratic fit models the row as a curve. The heading is taken as the look-ahead tangent and the cross-track as the lateral offset of that curve, and a temporal stage of a median filter, an exponential moving average, and a physical jump gate stabilises both signals. Because no component is trained, the field runs form a leakage-free test set, and the tracker emits a guidance line on every frame across a wide range of vegetation density. The evaluation is reported in navigation units, with cross-track also expressed in centimetres through a camera-to-ground homography built from the robot's measured camera pose; accuracy against human ground truth and a temporal-stage ablation are the subject of ongoing annotation. The contributions are a public forward-view cabbage-robot guidance dataset, a leakage-free in-field evaluation of a deliberately simple pipeline, and the hypothesis, which the evaluation is designed to test, that the temporal stage rather than the perception front end is the dominant contributor to heading stability. The study is bounded to one robot, one crop, one camera pose, and open-loop operation.

Keywords— crop-row guidance, vision-based navigation, vegetation index, robust line fitting, temporal filtering, agricultural robot, cabbage.

1. Introduction

Autonomous ground robots are increasingly used for the between-row work in precision agriculture that once required manual scouting and hand-steered passes. Their core requirement is to remain centred on the crop row across varying canopy and growth-stage conditions. A forward-view camera observes the canopy, a perception module localises the central row, and a *guidance line* summarising where the row leads is passed to a steering controller that keeps the vehicle centred [1–3]. On a low-cost robot without RTK-GPS this vision-only guidance line is the sole interface between perception and actuation, and what the controller consumes is a *heading* and a *cross-track offset*, not a detection score [4, 5].

Most crop-row systems obtain that line from a trained object detector, which raises two difficulties. The detector is optimised for a box-level objective such as mean average precision, a target that is not tied to the heading dominating the look-ahead, so it is tuned for the wrong quantity. It also needs field-specific labelled data that is costly to obtain and prone to leakage between training and test, and it introduces a learned failure surface that is hard to audit in the field [1, 5]. We therefore investigate whether crop-row guidance can be achieved without a trained detector.

We study a deliberately simple pipeline assembled entirely from classical computer vision and evaluate it in the field on a cabbage robot. An Excess-Green vegetation index [6, 7] and a column-peak tracker locate the central row; a robust quadratic fit $x = P(y)$ models its curvature [8, 9]; the heading is taken as the look-ahead tangent to the fitted curve and the cross-track as its lateral offset; and a temporal stage of a sliding-window median, an exponential moving average, and a physical jump gate stabilises both signals over time [10]. The pipeline has no learned weights and trains on nothing, so there is no train/test split to leak and the same code runs unchanged across crops and growth stages. All components are established; our aim is to quantify the guidance accuracy and temporal stability this stock composition achieves in field operation.

This paper makes three contributions: a real forward-view cabbage-robot crop-row guidance dataset, to be released publicly, recorded on a working field robot at a camera pose representative of low-cost in-row navigation; a leakage-free in-field evaluation of the detector-free, training-free pipeline, reported in the navigation units a controller consumes, namely heading in degrees and cross-track in pixels and in centimetres through a camera-to-ground homography built from the robot's measured camera pose, with accuracy against human ground truth the subject of ongoing annotation; and a testable hypothesis, which the evaluation is designed to assess, that the temporal stage rather than the perception front end is the primary source of robustness, examined by comparing heading error against ground truth on the hardest frames together with the gate's false-reject rate.

The scope is deliberately narrow. We study one robot, one crop, one camera pose, and open-loop operation, and we make no claim of generalisation beyond this setting. Section 3 describes the platform, the pipeline, and the evaluation protocol; Section 4 reports the results; and Sections 5 and 6 discuss the findings and conclude.

2. Related work

Four areas bound the present study. The first surveys vision-based crop-row guidance and the shift from classical pipelines to learned ones, and concludes that learned methods currently achieve the highest reported accuracy yet still depend on labelled data and remain exposed to domain shift. The second isolates the classical, handcrafted row-extraction recipe of vegetation index, thresholding, then a Hough or projection search, the family to which a training-free pipeline belongs. The third takes up an argument made with growing frequency in this literature, that crop-row methods ought to be measured in *navigation units* rather than detection mAP. Temporal filtering and path-tracking form the fourth, the back-end machinery that converts noisy per-frame geometry into a usable heading. Against this background sits a detector-free, temporally-robust pipeline validated on in-field cabbage video.

2.1. Vision-based crop-row guidance: classical versus learned

Steering a robot or tractor along a crop row from a forward-view camera is a long-standing task, and vision-guided field machinery goes back several decades [3, 11]. Early systems recover row geometry without any learned segmentation. Intensity-based crop-row determination by image analysis [5] and global energy minimisation over candidate row models [12] are representative of the classical, hand-engineered line of work.

Perception in this area has since moved decisively toward learned models. The de Silva et al. line of work segments the crop/soil mask with a network and recovers a centreline through a geometric scan, validated across many field conditions and growth stages [1, 2]. Detect-then-fit systems instead localise row patches or plant clusters with an object detector and fit a guidance line through the centroids, as in the YOLOX-based maize-row system of Gong et al. [13]. A further group regresses the row directly: RowDetr predicts each row as a polynomial [14], while a row–column attention model recasts the whole pipeline as end-to-end attention [15]. The row-line formulation itself descends from lane detection. There, Ultra-Fast structure-aware lane detection casts the problem as row-anchor classification [16], LaneATT performs anchor-based real-time detection [17], and Van Gansbeke et al. fit a lane as a differentiable least-squares line through detected points [18].

Learned methods currently achieve the highest reported accuracy, but at the cost of supervision. Each requires pixel- or row-level labels for the target crop, growth stage, and camera geometry, and its accuracy is assured only on distributions that resemble the training set. Degradation under changes of field, crop, and acquisition condition recurs across this family. For a single robot, a single crop, and a fixed camera pose, a training-free pipeline that needs no labels and has no parameters to overfit or leak is a label-free alternative warranting in-field validation against navigation-unit ground truth. The classical family that supplies such a pipeline is examined in the following subsection, and its robustness across a long real run remains unaddressed by the learned literature.

2.2. Classical, handcrafted row extraction

The pipeline deployed here belongs to the classical family that predates learned segmentation, and its components serve as stock parts rather than new contributions. Three stages make up the standard recipe. A vegetation index first separates plant from soil in colour space; the Excess-Green index and related colour indices for plant/weed discrimination under varied soil, residue, and lighting were established by Woebbecke et al. [6] and verified for automated crop imaging by Meyer and Neto [7]. The index image is then binarised, classically with Otsu's histogram threshold [19]. Row geometry is finally recovered from the resulting mask by a global search, either a Hough transform for real-time plant-row tracking [9] or the column projection and scan-line peaks used in vision-based row-following and row-detection systems for field machinery and sugar beet [20, 21].

Maturity, speed, and the absence of labels make these components a natural choice. The present pipeline instantiates the recipe with an Excess-Green column-peak central-row tracker. A robust quadratic fit $x = P(y)$ by Huber IRLS ($\delta = 1.345$, scale $1.4826\,\mathrm{MAD}$ [8]; RANSAC [22] is the standard robust alternative) replaces the single straight-line Hough or projection model, so that the guidance line can bow in the far field, and heading θ follows as the look-ahead tangent at a chosen row position, with cross-track offset e taken at the same point. The classical literature does not settle how such a single-frame extractor behaves over a long, real, forward-view field run across varying vegetation density, nor how its per-frame output should be stabilised in time. Both questions are addressed in the following subsection and in the experiments reported here.

2.3. Evaluation in navigation units

A recurring theme in recent crop-row work holds that detection mAP is the wrong yardstick for a guidance system. A detector can score well on mAP and still yield a poor guidance line, because what matters downstream is the geometry of the recovered row rather than the quality of individual boxes. The de Silva et al. benchmarks therefore report cross-track and heading error in degrees and pixels [1, 2], and the maize-row detect-then-fit system reports the angular error of the fitted centreline rather than box mAP [13]. Heading error and cross-track error, measured against a human-labelled guidance line, are the quantities that matter for guidance.

Navigation-unit evaluation needs a ground-truth line, which in turn needs human labelling and a study of inter-annotator agreement. Absent that line, an extractor can report how often it emitted a line but cannot report whether the emitted line was correct. The accuracy claim against a human ground truth and the inter-annotator agreement are therefore deferred [TBD pending GT labelling]. The camera calibration that expresses cross-track in centimetres is already available (Section 3.1).

2.4. Temporal filtering and path tracking

Per-frame row geometry from any single-frame extractor is noisy, so practical guidance systems stabilise it over time. The canonical tool is the Kalman filter, which fuses a motion model with noisy measurements for recursive state estimation [10] and underpins much lane- and row-tracking work. Once a stabilised heading and offset are available, classical path-tracking controllers convert them into steering. Pure pursuit follows a look-ahead point on the path [23], the Stanley controller couples heading and cross-track error for autonomous ground vehicles [24], and the kinematic bicycle model supplies the standard vehicle abstraction these controllers assume [25].

In the crop-row literature this back end is often treated as an implementation detail behind a learned front end, even though it is the dominant contributor to the per-frame stability of a guidance system. A transparent temporal stage, kept in front of a simple stock extractor, occupies the role that learned pipelines fill implicitly. The substantive robustness questions, namely heading error against a human ground truth on the frames where the raw extractor diverges, and the false-reject rate of the gate, require labels and are deferred [TBD pending GT labelling], together with baseline comparisons against RANSAC and equal-weight least-squares fits and an ablation of the temporal stages.

2.5. Positioning of the present study

Where the literature trends toward learned perception that needs labels and is exposed to domain shift, the gap left open is a detector-free and training-free pipeline assembled from stock classical parts: an Excess-Green column-peak tracker [6, 7, 19], a robust quadratic fit $x = P(y)$ by Huber IRLS [8], a look-ahead tangent heading, and a median/EMA/jump-gate temporal stage [10]. All components are established. Positioned against prior art, the open question is whether such a pipeline holds up on real forward-view cabbage field video and can be validated in navigation units against a human ground truth [TBD pending GT labelling], a question the remaining sections take up.

3. Materials and methods

This work is a validation and deployment contribution rather than a new method. No new perception architecture, line estimator, or controller is proposed. The guidance pipeline is assembled entirely from established, stock computer-vision components: an Excess-Green index for vegetation segmentation, a robust quadratic line fit, a look-ahead tangent heading, and temporal smoothing by a median filter, an exponential moving average, and a jump gate. The whole pipeline runs in plain `cv2/numpy` with no neural detector and no training of any kind. The contribution is an in-field validation of this detector-free and training-free pipeline on a working cabbage-field robot. Three results follow from that validation. The first is a public real forward-view cabbage robot dataset. A leakage-free evaluation tied to human ground truth and expressed in navigation units forms the second. The third is the empirical finding that the temporal stage, and not the perception front end, is the primary source of robustness, demonstrated as error against ground truth and as a false-reject rate rather than as a jump-count reduction that follows by construction from the gate threshold. Since the pipeline never trains on any frame, no data can leak from a training set into the evaluation, and every frame is a test frame.

Several quantitative results require human-labelled ground truth that is still being produced. To keep the manuscript auditable, any quantity that cannot yet be reported from data is written explicitly as "[TBD pending GT labelling]". The quantities reported as measured in this paper are the row-emission rates and the descriptive jump counts of Section 3.5, which are labelled as such; the remaining quantities are pending.

3.1. Robot platform and field data

The data were collected from a single ground robot (Figure 1), a four-wheel straddle platform that drives along a crop row with a forward-view camera mounted to look down the inter-row corridor. The robot was driven between cabbage rows under manual/teleoperated control at a nominal forward speed of approximately $0.14\,\mathrm{m\,s^{-1}}$. The platform pairs an embedded CPU-only computer and a forward-view USB webcam at 640×480, here a Logitech HD C270 webcam and an embedded board of the NVIDIA Jetson Nano / Raspberry Pi 4 class. The detector-free pipeline of this paper runs on its CPU with no neural accelerator (Figure 2). This robot shares its camera model with a sibling robot whose camera was calibrated from a standard checkerboard procedure; the intrinsics and lens distortion from that calibration are reused here, as they are a property of the camera and lens rather than of the mounting. The extrinsics, by contrast, are taken from this robot's own measured installation: the camera sits on the robot centreline 65 cm above the ground with its optical axis tilted $55°$ from the vertical (equivalently $35°$ below the horizontal). From these intrinsics and

Figure 1: The cabbage-field robot during a field-test session: a four-wheel straddle platform carrying a forward-view camera, operated between cabbage rows under manual/teleoperated control. All data in this study were recorded this way; closed-loop autonomous guidance from the proposed pipeline is left to future work.

measured extrinsics an image-to-ground homography is constructed under a flat-ground assumption (world origin on the centreline beneath the camera) and used to express the near-row cross-track in centimetres (Section 3.4). The residual assumptions are the planar ground and the reuse of the camera intrinsics, so the cross-track is also reported in pixels and as a fraction of the half-image width. The scope is fixed to one robot, one crop (cabbage), and one camera pose. The validation reported here is narrow by design, open-loop, and at a single low frame rate, with no claim of generalisation beyond this configuration. Closed-loop autonomous steering on this platform is future work and was not exercised.

The full sensing-to-actuation hardware path appears in Figure 3. The forward-view camera streams

Figure 2: Onboard camera and compute. The forward-view camera and the embedded computer are carried on the four-wheel straddle frame that spans the crop bed. The bed shown here is mixed, with cabbage interplanted among other leafy greens, as also reflected in the field runs (Section 4.1).

over USB to the onboard computer, which runs the detector-free guidance pipeline and emits a heading and a cross-track offset. A microcontroller and stepper drives turn the wheels. In the present study only the perception path was run and logged, from camera to guidance output; the steering controller and the closed loop to the drive were not exercised, and they are marked as future work. Two real robot runs recorded on cabbage plots form the frame set: run 1 (approximately 20 minutes) and run 2 (approximately 14 minutes). Frames are sampled at 1 frame per second (1 fps) and screened to the frames showing the cabbage navigation row through the look-ahead region, removing stretches dominated by adjacent leafy-green beds or by occluding operators. This yields 1482 forward-view cabbage frames in total (812 from run 1 and 670 from run 2), all at 640×480. The full set is treated as a pure test set. Because the pipeline is training-free there is no training/validation

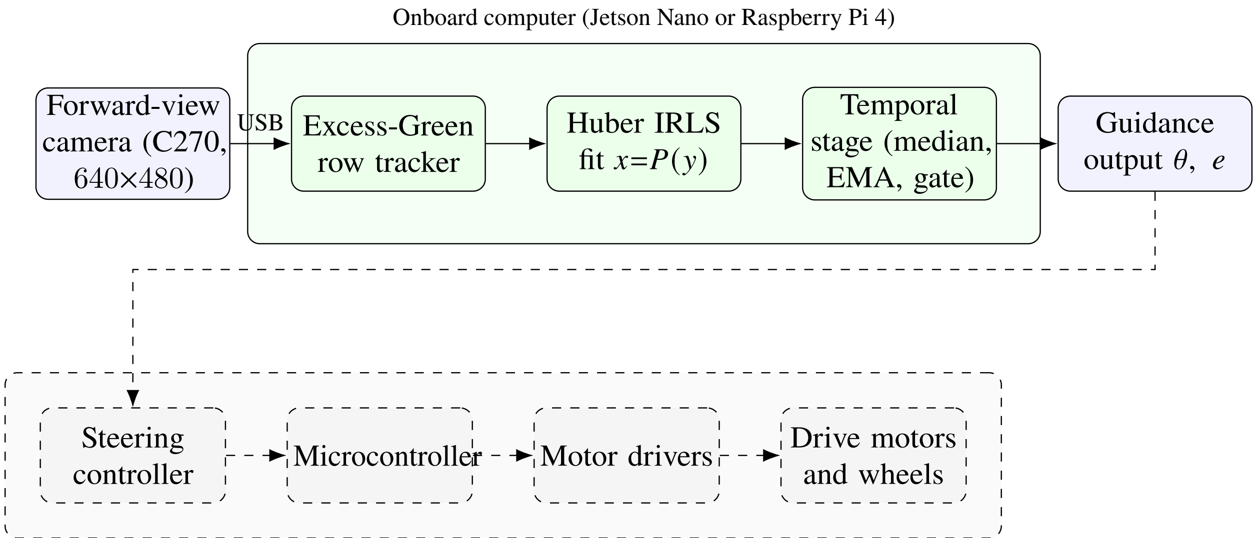

Figure 3: Hardware and data path of the cabbage robot. The solid blocks, sensing and the onboard detector-free guidance pipeline, were run and logged in this study; the dashed group (steering controller, microcontroller, stepper drivers, and motors) was not exercised here and shows the intended closed loop.

split and no possibility of memorisation. To probe robustness to crop growth stage and canopy density the frames are stratified into vegetation tertiles (low, mid, and high Excess-Green coverage), giving three roughly equal strata of 491, 487, and 504 frames. The dataset will be released publicly as the first real forward-view cabbage robot crop-row corpus, and a Data and Code Availability statement accompanies the manuscript.

3.2. Central-row tracking via the Excess-Green index

Each frame is reduced to a single-channel vegetation map using the Excess-Green index of Woebbecke *et al.* [6]. From the normalised RGB channels (r, g, b) the index is

$$\mathrm{ExG} = 2g - r - b, \tag{1}$$

which emphasises live green canopy against soil and residue and is a standard, illumination-tolerant choice for in-field crop/soil separation [7]. The map is thresholded into a binary vegetation mask by Otsu's method [19], which selects the foreground/background split automatically for each frame and so adapts to changing lighting and canopy density without a hand-tuned constant.

Crop rows viewed head-on appear as near-vertical green bands, and the central navigation row is the band the robot straddles. Tracking it without a detector relies on a per-row column profile. For each image row y the pipeline sums (or smooth-and-sums) the masked ExG response across columns within a central corridor of the image, and takes the column of peak vegetation density as the candidate row centre $x^*(y)$ at that height. Scanning y from the image bottom upward yields a set of central-row anchor points $\{(x^*(y), y)\}$ that follow the near-vertical green band. Restricting the column search to a central corridor, and seeding it at the bottom-centre of the frame where the navigation row is closest to the robot, biases the tracker toward the row the robot is actually in rather than an adjacent row. The output of this stage is the ordered set of central-row anchors passed to the line fit. Emitting an anchor set is a self-consistency property and not a correctness guarantee: a peak is found in essentially every frame, but whether the chosen column corresponds to the true central row is precisely what the ground-truth evaluation of Section 3.7 is designed to measure.

3.3. Robust quadratic guidance-curve fitting

The guidance line is parameterised as image column x as a function of image row y,

$$x = P(y) = c_2\, y^2 + c_1\, y + c_0, \tag{2}$$

a quadratic in y. This orientation is chosen for a reason. Crop rows viewed head-on are near-vertical, so a conventional $y = mx + c$ fit becomes ill-conditioned as the slope diverges; expressing x as a function of y removes this vertical-line degeneracy [5]. The quadratic term c_2 captures the gentle far-field bowing of a row under perspective and mild row curvature, and reduces to the straight-line case $x = c_1\,y + c_0$ in the limit $c_2 \to 0$. The instantaneous local heading at any height follows from the slope $P'(y) = 2c_2 y + c_1$,

$$\theta(y) = \arctan\bigl(P'(y)\bigr), \tag{3}$$

reported in degrees from the vertical as the controller-independent navigation quantity; its evaluation at a look-ahead height is detailed in Section 3.4.

Given the n central-row anchors $\{(x_i, y_i)\}$ from Section 3.2, the coefficients $\mathbf{c} = (c_2, c_1, c_0)$ are obtained by a standard robust quadratic fit $x = P(y)$ by Huber iteratively-reweighted-least-squares (IRLS) regression of x on y [8], minimising

$$\hat{\mathbf{c}} = \arg\min_{\mathbf{c}} \sum_{i=1}^{n} \rho_\delta\bigl(x_i - P(y_i)\bigr), \qquad \rho_\delta(r) = \begin{cases} \frac{1}{2}r^2, & |r| \le \delta, \\ \delta\bigl(|r| - \frac{1}{2}\delta\bigr), & |r| > \delta. \end{cases} \tag{4}$$

The coefficients are estimated by standard Huber IRLS with $\delta = 1.345$ and a MAD scale $\hat{\sigma} = 1.4826\ \mathrm{median}_i\,|r_i|$, with a small ridge for conditioning, giving the Huber influence weights

$$w_i = \min\left(1,\ \frac{\delta\,\hat{\sigma}}{|r_i|}\right), \qquad \delta = 1.345. \tag{5}$$

The constant $\delta = 1.345$ is the conventional Huber tuning that retains $\approx 95\%$ efficiency under Gaussian noise while bounding the influence of outliers, which here are spurious anchors from gaps, weeds, or an adjacent row.

Two standard alternatives serve as comparison fits over the same anchor set, evaluated against the deployed Huber IRLS fit: equal-weight least squares (the non-robust polynomial fit) and RANSAC consensus fitting [22], a strong robust baseline. The accuracy of the deployed fit relative to these baselines against human ground truth is reported in Section 4, and those comparison numbers are [TBD pending GT labelling].

3.4. Heading and cross-track geometry

A controller steers not to the row directly beneath the robot but to where the row is heading some distance ahead. The guidance geometry is therefore evaluated at a fixed look-ahead image row y_{la} in the upper portion of the frame, mirroring the look-ahead point of pure-pursuit and Stanley-type path trackers [23–25]. Because the curve $P(y)$ bows in the far field, the relevant heading is the tangent to the curve at the look-ahead height, not a chord to the image bottom. The reported heading is therefore the look-ahead tangent

$$\theta = \arctan\big(P'(y_{\mathrm{la}})\big) = \arctan\big(2c_2\, y_{\mathrm{la}} + c_1\big), \tag{6}$$

the slope of Equation (3) evaluated at y_{la}, in degrees from the vertical. Using the tangent rather than a global slope aligns the reported heading with the far-field row direction the robot must follow, and avoids the bias a straight-line fit would incur on a bowed row.

The lateral cross-track offset e is the signed horizontal distance between the robot's reference column, taken as the image vertical centreline $x_c = W/2$ for a forward-view camera with $W = 640$, and the guidance line evaluated at the look-ahead row,

$$e = P(y_{\mathrm{la}}) - x_c, \tag{7}$$

in pixels. The offset e is additionally reported normalised by the half-image width $W/2$ (that is, as a fraction of the half-frame), so that it is dimensionless and comparable across mountings. Conversion to centimetres requires a metric ground anchor, namely an image-to-ground homography under a flat-ground assumption. This anchor is the homography of Section 3.1, built from the robot's measured camera height and tilt; the centimetre cross-track magnitude follows directly, while e is in all cases also reported in pixels and in % of half-image width.

3.5. Temporal filtering

The per-frame heading θ and offset e are smoothed over time by three stock, lightweight operators applied in sequence. Together they form the temporal stage, which is the primary source of robustness. The chain is far simpler than a full state estimator such as a Kalman filter [10]; its complexity is matched to the 1 fps, open-loop operating regime.

The first operator passes the raw heading stream through a short sliding-window median filter. The median is robust to isolated single-frame outliers, such as a momentary wrong-row peak or a gap-induced bad fit, without the lag or overshoot of a mean filter. The median-filtered heading is then low-pass filtered by an exponential moving average,

$$\hat{\theta}_t = \beta\,\hat{\theta}_{t-1} + (1-\beta)\,\tilde{\theta}_t, \tag{8}$$

where $\tilde{\theta}_t$ is the median-filtered heading at frame t and $\beta \in [0,1)$ sets the smoothing horizon. The exponential moving average suppresses high-frequency jitter while remaining causal and $O(1)$ in memory. The offset e is fused with the identical median→EMA chain.

A frame-to-frame jump gate completes the chain, rejecting any update whose heading change exceeds a physical bound $\Delta\theta_{\max}$. When $|\tilde{\theta}_t - \hat{\theta}_{t-1}| > \Delta\theta_{\max}$ the update is gated and the last fused heading is held rather than applied. The bound $\Delta\theta_{\max}$ is set from the robot's physical heading-rate limit, independently of any evaluation threshold. A ground robot driving a row cannot reorient faster than some maximum yaw rate $\dot{\psi}_{\max}$ (deg s^{-1}), so over one inter-frame interval Δt the largest kinematically plausible heading change is

$$\Delta\theta_{\max} = \dot{\psi}_{\max}\,\Delta t, \qquad \Delta t = \frac{1}{1\,\text{fps}} = 1\,\text{s}. \tag{9}$$

Any apparent inter-frame heading change larger than $\Delta\theta_{\max}$ cannot be the vehicle physically turning; it has to be a perception artefact, such as a jump onto an adjacent row, and is gated. The value $\dot{\psi}_{\max}$ is fixed from the platform's turning specification, and hence $\Delta\theta_{\max}$, before any error metric is examined, so the gate threshold is not tuned to the evaluation. The same physical reasoning bounds the per-frame change in e.

A reduction in large per-frame jumps is not the measure of the gate's effect, since the gate rejects large jumps by definition and any "before/after" jump count is therefore arithmetic rather than evidence of correctness. Raw-versus-fused jump counts are reported descriptively in Section 4, labelled as such, and are excluded from the robustness results. The substantive robustness claim, that fusion lowers heading and cross-track error against human ground truth (including on the large raw-jump frames) at an acceptable false-reject (gate) rate, is evaluated in Section 4 and is [TBD pending GT labelling].

Figure 4 shows the complete pipeline applied to a single forward-view cabbage frame, from the Excess-Green track through the robust curve to the look-ahead tangent and the reported heading and cross-track.

3.6. Ground-truth annotation protocol

The ground-truth (GT) guidance line used for evaluation is human-annotated and is therefore independent of the perception pipeline under test. For each labelled cabbage frame an annotator

Figure 4: Detector-free guidance on a real forward-view cabbage frame. *Left*: input frame (640×480). *Right*: the Excess-Green column-peak track (orange points), the robust Huber IRLS quadratic guidance curve $x = P(y)$ (green), the look-ahead tangent that defines the steering heading θ (red), and the near-row cross-track offset e from the image centre (blue); the reported θ and e are overlaid. No detector and no training are involved.

marks the central navigation row directly, following *convention A*: the line is labelled along the row the robot is driving, traced through the cabbage canopy from the image bottom-centre up the inter-row corridor. The marked points are fit in the same $x = P(y)$ parameterisation of Equation (2), and the GT heading is the look-ahead tangent of Equation (6) evaluated at the same y_{la} as the system output, so that predicted and GT headings are directly comparable. A subset of frames is double-labelled to quantify the reliability of this protocol, with inter-annotator agreement reported as the median absolute heading difference in degrees and cross-track in pixels; these agreement figures are [TBD pending GT labelling]. Because the GT line is marked by hand on the image rather than derived from the tracker's anchors, it is independent of the Excess-Green front end and provides a fair external reference. The labelling effort is in progress, so the agreement figures are reported as pending.

3.7. Evaluation metrics and statistics

The primary metric is heading error in degrees, the absolute difference between the system's look-ahead tangent heading and the human-GT heading, $|\theta - \theta^{\mathrm{gt}}|$, which is controller-independent. The secondary metric is cross-track error in pixels (and in % of half-image width), the lateral offset between the predicted and GT guidance lines at the look-ahead row y_{la} (Equation (7)). Centimetre units are reported through the image-to-ground homography of Section 3.1, built from the robot's measured camera pose; the cm-scale errors await the GT subset and are [TBD pending GT labelling], and in all cases the errors are reported in degrees and pixels.

As a basic sanity check the fraction of frames for which the tracker emits a guidance line at all is reported as a self-consistency measure, both overall and broken down by vegetation tertile; the measured value is reported in Section 4. Here "row found" means a line was emitted, not that the emitted line is the correct central row. Whether the emitted line is correct is measured by the heading and cross-track error against human GT above, which is [TBD pending GT labelling].

Consistent with Section 3.5, the robustness of the temporal stage is reported as (i) heading and cross-track error versus human GT stratified by raw front end difficulty, in particular on the large raw-jump frames, and (ii) the false-reject (gate) rate, that is, the fraction of frames the physical jump gate holds. The raw-versus-fused jump count is excluded from the robustness results. Both (i) and (ii), together with the deployed-fit-versus-baseline (equal-LS, RANSAC) and temporal-ablation comparisons, are [TBD pending GT labelling].

Per-frame errors are written to one CSV row per frame so that the deployed pipeline and its baselines and ablations are compared paired on identical frames. Because per-frame errors are not normally distributed and frames within a run are temporally correlated, significance is assessed with paired non-parametric tests, specifically the Wilcoxon signed-rank test on paired per-frame differences, with Holm–Bonferroni correction across the confirmatory comparison family. Effect sizes (Cliff's δ) are reported with bootstrap confidence intervals rather than relying on p-values alone. Any comparison whose median difference is smaller than its dispersion is downgraded in wording. All confirmatory accuracy numbers are [TBD pending GT labelling].

4. Results

This section reports what two real cabbage-field robot runs allow us to establish, separating what those runs *measure* from what still depends on hand-labelled ground truth (GT). The pipeline under test is the detector-free and training-free guidance stack of Section 3: an Excess-Green index [6, 7] column-peak central-row tracker, a robust quadratic fit $x = P(y)$ by Huber IRLS [8], a look-ahead tangent that defines the heading θ together with the cross-track offset e, and a temporal stage built from a sliding-window median, an EMA, and a physical max-jump gate. The system is never trained, so the two runs serve as a pure test set. No parameters are fitted on these frames, which removes any path for leakage.

Two boundaries govern every number that follows. Self-consistency is not correctness: the "row-found" rate of Section 4.2 records only that the tracker *emitted* a guidance line for a frame, and it carries no claim about whether that line is the correct central row. The jump counts are likewise descriptive rather than a robustness result. Because the temporal stage suppresses large frame-to-frame heading jumps as a direct effect of the gate threshold, the post-fusion jump counts in Section 4.3 follow arithmetically from that threshold and do not show that the fused heading is more accurate. The robustness claim that actually matters, heading error against human GT on the very frames the gate suppresses, paired with the false-reject rate, is *[TBD pending GT labelling]* and is marked as such in every table. Cross-track is reported in pixels, as a percentage of the half-image width, and in centimetres through the image-to-ground homography of Section 3.1, built from the robot's measured camera height and tilt.

4.1. Dataset

The evaluation data are two forward-view video runs recorded by a forward-view camera on a cabbage field robot, sampled at 1 fps and screened to 1482 forward-view cabbage still frames at 640×480. The two runs cover different sections of the field at different times of day, so vegetation density and lighting differ between them. Table 1 summarises the runs. The setting is one robot, one crop, one camera pose, and open-loop logging at 1 fps, and the dataset supports no claim of coverage beyond this narrow operating envelope.

Table 1: Field dataset: two real forward-view cabbage robot runs used as a pure test set (no training, so no leakage). Frames are sampled at 1 fps.

Run	**Duration**	**Frames**	**Rate (fps)**	**Resolution**	**Camera**
Run 1	~20 min	812	1	640×480	forward-view
Run 2	~14 min	670	1	640×480	forward-view
Total	~34 min	**1482**	1	640×480	forward-view

4.2. Row-emission reliability

We first assess whether the perception front end emits a guidance line on every frame. Across all 1482 frames the Excess-Green index column-peak tracker emitted a guidance line in 100% of frames ($1482/1482$). To test whether this figure conceals a silent failure mode in sparse or dense canopy, we split the frames into vegetation tertiles by per-frame Excess-Green coverage (low, mid, high). Table 2 shows that the emit rate is 100% in every tertile. This stratification is partly tautological by construction: the tracker locates the strongest Excess-Green column, so a peak exists whenever any green is present, and emission across Excess-Green-defined tertiles is close to guaranteed by the design of the front end. The result therefore shows only that the tracker never aborts. Whether the canopy density leaves the row correct is the separate accuracy question of Section 4.4.

The 100% row-found rate is a self-consistency statement: in every frame, including the low-vegetation frames where the canopy signal is weakest, the pipeline returned a line rather than aborting. Figure 5 overlays the fitted guidance line on four representative cabbage frames. The rate is not a correctness statement, since it does not show that the emitted line is the correct central crop row, only that a line was produced. Whether those lines are right is an accuracy question that requires human GT, reported (pending) in Section 4.4.

Table 2: Reliability (self-consistency, *not* correctness): fraction of frames in which the tracker emitted a guidance line, overall and by vegetation tertile (Excess-Green coverage). A 100% rate means a line was *emitted*, not that it is the correct central row.

Stratum	**Frames**	**Row emitted**	**Rate**
All frames	1482	1482	100%
Vegetation: low (T1)	491	491	100%
Vegetation: mid (T2)	487	487	100%
Vegetation: high (T3)	504	504	100%

Mapping the per-frame near-row pixel offset through the image-to-ground homography (Section 3.1) describes the magnitude of the lateral offset in metric units. The central row sat a median of 1.9 cm from the image centre across all 1482 frames (mean 5.3 cm, 90th percentile 16.3 cm), where the centimetre conversion uses the homography built from the robot's measured camera pose (Section 3.1) and is reported alongside the pixel and percentage-of-half-width values. These figures characterise the *magnitude* of the lateral offset the controller would have observed, a geometric summary of the runs rather than an error against ground truth. Whether the estimated offset is *correct* is the separate, pending accuracy question of Section 4.4.

Figure 5: Qualitative behaviour of the detector-free estimator on four real forward-view *cabbage* frames from the two runs (each panel overlays the Excess-Green column-peak track in orange, the robust quadratic guidance curve $x = P(y)$ in green, the look-ahead tangent/heading in red, and the near-row cross-track in blue; the reported θ and e are printed). The tracker follows the central cabbage row across growth stages and lighting; the top-left frame also shows operators standing beside the robot during the supervised run. These panels are illustrative, not a correctness measurement (cf. Section 4.4).

4.3. Temporal stability

The temporal stage, a sliding-window median followed by an EMA and a physical max-jump gate in the spirit of standard recursive smoothing [10], exists to suppress single-frame heading excursions that a controller should not act on. Table 3 reports how many per-frame heading jumps exceed $10°$, $15°$, and $20°$ in the raw per-frame tracker output and after the temporal stage. The per-frame jump is the change in heading from the previous frame and is undefined for the first frame of each run, so the denominators are 812 and 670 as logged.

These counts are descriptive, since the gate suppresses the jumps by design. The post-fusion counts drop to nearly zero because the max-jump gate rejects exactly these large frame-to-frame changes. That reduction is an arithmetic property of the gate threshold rather than an independent measurement of correctness, and it does not amount to "jumps eliminated." The question that decides whether the temporal stage helps, namely whether the fused heading is closer to human GT than the raw heading on the suppressed frames and how often the gate rejects a correct large heading, is tested in Section 4.4 and is *[TBD pending GT labelling]*.

Table 3: Per-frame heading-jump counts (change from the previous frame) exceeding $10°/15°/20°$, raw versus after the temporal stage, as exact counts on the logged frames. The post-fusion reduction follows arithmetically from the max-jump gate and is not a correctness result; the correctness test on these frames is reported in Section 4.4.

Run / stage	$> 10°$	$> 15°$	$> 20°$
Run 1 raw ($n = 812$)	102	58	35
Run 1 temporal stage	0	0	0
Run 2 raw ($n = 670$)	194	130	89
Run 2 temporal stage	2	0	0

4.4. Accuracy and ablation evaluation

The quantities on which the deployment claim rests are the accuracy against a human labelling of the central row and the comparisons that interpret it. Each requires the hand-labelled GT subset, in which an annotator marks the central navigation row per frame and the GT line and its look-ahead tangent are derived in the same $x = P(y)$ parameterisation; inter-annotator agreement then sets a floor on the smallest meaningful error. That labelling is in progress, so every quantity in Table 4 is *[TBD pending GT labelling]*, with the metrics fixed in advance to keep the evaluation confirmatory. Accuracy is the heading error $|\theta - \theta^{\mathrm{GT}}|$ in degrees and the cross-track error $|e - e^{\mathrm{GT}}|$ in pixels, as a percentage of the half-image width, and in centimetres through the homography of Section 3.1, each summarised by median, mean, and 90th percentile. The temporal-stage question of Section 4.3 is tested here as the heading error against GT on the suppressed large raw-jump frames, raw versus fused, together with the gate's false-reject rate. The deployed fit is compared against a RANSAC line fit [22] and an equal-weight least-squares fit over the same Excess-Green column peaks, and the temporal stage is ablated (raw versus median+EMA+gate); a self-consistency or jump-count comparison would not separate these arms, so the discriminating axis is error against the labelled central row.

Table 4: Pre-registered accuracy and ablation evaluation against human ground truth, consolidating the pending quantities into one scaffold. Every entry is *[TBD pending GT labelling]* until the hand-labelled subset is complete; the metrics and comparison arms are fixed in advance.

Quantity (versus human ground truth)	**Status**
Heading error $\lvert\theta - \theta^{\mathrm{GT}}\rvert$ (median / mean / p90, °)	[TBD pending GT labelling]
Cross-track error (px; % half-width; cm)	[TBD pending GT labelling]
Inter-annotator agreement (heading °; cross-track px)	[TBD pending GT labelling]
Heading error on large raw-jump frames, raw vs. fused (°)	[TBD pending GT labelling]
False-reject rate of the max-jump gate	[TBD pending GT labelling]
Fit arm: quadratic $x{=}P(y)$ vs. RANSAC vs. equal-weight LS	[TBD pending GT labelling]
Temporal ablation: raw vs. median+EMA+gate	[TBD pending GT labelling]

The two runs establish that the front end is reliable in the self-consistency sense, with a line emitted in 100% of 1482 frames including the low-vegetation tertile, and they give descriptive context on how often the raw per-frame heading would have jumped. Accuracy is not yet established.

Every correctness quantity remains *[TBD pending GT labelling]*: heading and cross-track error against human GT, inter-annotator agreement, error against GT on the gate-suppressed large raw-jump frames, the false-reject rate, and the baseline and temporal-ablation comparisons of Table 4. Completing the hand-labelled subset is the one remaining step that turns this from a self-consistency report into the accuracy validation the deployment claim requires.

5. Discussion

This work is an applied contribution: an in-field validation of a detector-free and training-free crop-row guidance pipeline on a cabbage robot, rather than a methods breakthrough. Every component is a stock part. The front end uses the Excess-Green index for vegetation contrast [6, 7], a robust quadratic fit $x = P(y)$ by Huber IRLS [8, 22], and a look-ahead heading that is taken as the tangent to the fitted guidance line; the temporal stage is a sliding-window median, an EMA, and a max-jump gate [10]. All components are established. The contribution is that, assembled in this order and validated on real forward-view cabbage video, the combination yields a deployable vision-only guidance signal, and that the temporal stage, rather than the perception front end, is the primary source of robustness. The discussion is organised around that thesis: the conditions under which it holds, and the limits of the evidence behind it. Throughout, the quantities that the two runs already measure are kept separate from those that remain pending ground-truth (GT) labelling.

5.1. Large-error suppression as the safety-relevant quantity

The heading θ, defined as the look-ahead tangent to the fitted guidance line $x = P(y)$, is the wrong interface to a steering controller if it is reported as a bare number, however accurate that number is on average. A controller integrates whatever heading it is handed, so the quantity that decides whether the robot stays between the rows or drives into a stand of cabbage is not the *mean* heading error but the rate of *large* heading errors that reach the actuator. A single frame in which the Excess-Green front end latches onto a weed, a gap, or the wrong row can swing the look-ahead tangent by tens of degrees, and one such residual spike, if acted on, can damage plants and force a human intervention. Whereas a typical heading error produces a slightly off-centre pass, a single large residual error can cause the robot to damage plants and require human intervention. Suppressing large-heading-error frames therefore matters more than shaving the mean, and that is exactly what a temporal stage with a physical max-jump gate is built to remove.

The per-frame jump counts of Section 4 provide context rather than a headline result. As reported in Table 3, raw headings show frequent $> 10°$ frame-to-frame jumps on both runs, and these are largely eliminated after fusion. Because the max-jump gate rejects large frame-to-frame swings as a direct consequence of its threshold, the post-fusion count of large jumps is an arithmetic property of the gate rather than independent evidence that the emitted heading is correct. The safety claim that matters concerns error against human GT on exactly those spike frames, that is, whether the fused heading the robot would have steered on is close to a human's intended heading when the raw front end spiked. That number is *[TBD pending GT labelling]*. Until those labels exist, the jump counts establish only that the front end is jumpy and the temporal stage is smooth, which is necessary but not sufficient for safety.

5.2. The hypothesised role of the temporal stage

The working hypothesis is that the front end perception choice is secondary, and that the temporal stage is the dominant robustness lever. The labelled evaluation below is built to test that proposition; the argument here states why the hypothesis is expected to hold, not that the runs have already shown it, and it follows the same failure mode as above. The Excess-Green column-peak tracker is a single-frame estimator. It commits, per frame, to whichever column of vegetation contrast is strongest, with no memory and no sense of how plausible that commitment is given the last few seconds of driving. Swapping it for a different stock front end (a segmentation scan, a row-anchor estimator, or another vegetation index) should not change the underlying problem, because the single-frame estimate will still occasionally lock onto a weed patch, a canopy gap, or an adjacent row. Such errors arise from scene content rather than from the estimator, and persist under any single-frame front end. The temporal stage is what is expected to convert an occasionally-wrong per-frame estimate into a heading a controller can integrate. The sliding-window median rejects isolated contaminated frames, the EMA limits how fast the steering target can move, and the max-jump gate enforces that the look-ahead heading cannot change faster than the vehicle kinematically can. If the dominant errors are sparse-in-time spikes, a median-plus-gate should remove them regardless of which front end produced them, which is why the lever is expected to be temporal.

The measurements needed to test the hypothesis remain to be collected. Proper evidence is a temporal ablation measured against human GT (front end alone versus front end plus fusion, scored as heading error and, above all, as the rate of large-error frames that survive to the output), together with a comparison against detect-then-fit and RANSAC / equal-least-squares baselines under the same temporal stage. Those comparisons are *[TBD pending GT labelling]*. The descriptive jump statistics are consistent with the hypothesis but cannot stand in for it (see Section 5.1): a temporal stage that suppresses jumps by also suppressing *correct* large corrections would smooth the trace while raising error against GT, and only the labelled comparison can separate the two cases.

5.3. Safety conditions for the max-jump gate

The max-jump gate is the safety component of the pipeline only under a condition that remains unmeasured, namely that its *false-reject rate* is low. The gate's job is to veto a proposed heading update that is physically implausible. It is genuinely protective when the vetoed updates are the contaminated ones, the weed-lock and wrong-row spikes, because it then holds the last trustworthy heading through a transient and resumes when the front end recovers. The same gate, with the same threshold, will also veto a *real* sharp correction: at a true headland, at a sharp bend in the rows, or after the robot has genuinely drifted and the front end is now reporting the correct large heading needed to recover. Rejecting those updates does not add safety; it suppresses the very correction that keeps the robot in the lane, and it does so silently. The distinguishing measurement is the false-reject rate: among the frames the gate vetoes, what fraction were actually correct large headings (against human GT) rather than contamination. That rate is *[TBD pending GT labelling]*, and the gate can be called a safety mechanism only once it is shown to be low. As noted in Section 5.1, suppressing large jumps is an arithmetic property of the gate, so only the false-reject number tells us whether it is rejecting the right ones.

5.4. Failure modes of the Excess-Green front end

Because the temporal stage can only stabilise what the front end gives it, the failure modes of the Excess-Green tracker bound the system. They are the familiar limits of vegetation-index row sensing [6, 7, 9], and they bear on cabbage forward-view imagery in specific ways. *Low vegetation contrast:* the Excess-Green index separates green canopy from soil by colour, so at early growth stages, over wet or dark soil that is itself greenish, or under flat overcast light, the column-peak signal is weak and the tracked centre wanders. Reporting that a row was found in 100% of frames (Section 4) does not contradict this. "Found" means a line was emitted, not that the emitted line is the correct row; a low-contrast frame still yields a column peak, just a less reliable one, and whether that peak sits on the right row is a question only GT can answer. *Weeds:* inter-row weeds are also green and also produce Excess-Green columns, so a weed band between rows can out-peak the crop row and pull the tracker off-centre, a content failure that no amount of robust fitting on a single frame can detect. *Canopy closure:* as cabbage matures and neighbouring rows touch, the inter-row soil gap that defines the column structure closes, the peaks merge, and the central-row column becomes ambiguous; the tracker may then follow the seam between two rows rather than a row centre. *Gaps and missing plants:* a stretch of missing plants removes vegetation contrast locally, so the look-ahead portion of the curve is fit on little or no support and the tangent becomes unstable exactly where heading is most sensitive. In all four cases the front end produces a confident-looking but wrong per-frame line, which is why the temporal stage matters. It is also why the temporal stage's protection is bounded by its false-reject rate: a sustained content failure such as a long weed band or a closed-canopy stretch is not a transient spike, and a median-plus-gate designed for spikes will eventually track it.

5.5. Limitations

Several boundaries limit how far these conclusions generalise. The evaluation is open-loop: recorded forward-view runs are processed offline, with no robot in the loop, no wheel odometry, and no actuation. A closed-loop intervention rate and realised tracking error are therefore out of reach, and the smoothness of the fused heading trace is not the same thing as closed-loop stability. The temporal parameters (window length, EMA factor, jump threshold) interact with controller and vehicle dynamics that an offline replay does not capture. A closed-loop field trial with odometry is the clear next step.

The metric figures rest on a measured camera pose and a flat-ground assumption. The heading θ is reported in degrees, which is controller-relevant and scale-free, and the cross-track offset e is reported in pixels, as a fraction of half the image width, and in centimetres. The centimetre conversion uses an image-to-ground homography built from the robot's own measured camera height (65 cm) and tilt (55° from the vertical), with the camera intrinsics reused from a checkerboard calibration of the same camera. A flat-ground homography still inflates metric error most at the far look-ahead distance, precisely where guidance is most sensitive, so the centimetre figures away from the near row carry that planar-ground caveat. A full on-robot checkerboard calibration, including the intrinsics, would further tighten the metric mapping and is a clean refinement.

The scope is narrow: two runs, one crop, one camera pose. The evidence is two real cabbage runs (~ 20 and ~ 14 minutes, 1482 curated cabbage frames at 1 fps, 640×480), from one robot, one crop, and one forward-view camera pose. This is a pure test set, since there is no training stage anywhere in the pipeline and nothing can leak from these frames into the method, but it is also a small, single-condition sample. With $n = 2$ runs the across-field, across-season, and across-platform variability cannot be characterised, and the descriptive front end statistics already differ markedly between the two runs (raw $> 10^\circ$ jumps in 13.9% versus 31.8% of frames), which is itself a caution against generalising from two clips.

Self-consistency is not correctness, and the headline numbers are still pending. The two facts that can be stated without labels (a line emitted in 100% of frames, in every vegetation tertile, and the collapse of large frame-to-frame jumps after fusion) are *self-consistency* properties of the pipeline, not *correctness* properties. “Row found” means the tracker emitted a line, not that the line is on the correct row. “Jumps suppressed” means the gate did its arithmetic job, not that the surviving heading is right (Section 5.1). The quantities that would make this a correctness result are heading error against human GT, cross-track error, error against GT on the spike frames, the gate’s false-reject rate, inter-annotator agreement, and the temporal-ablation and baseline comparisons, all of which are *[TBD pending GT labelling]*. The current results stand as a leakage-free in-field validation of feasibility and of front end / temporal behaviour, and every accuracy and safety claim is deferred to the labelled evaluation.

6. Conclusions

On a low-cost field robot without RTK-GPS, the quantities a steering controller actually consumes are a heading and a cross-track offset, not a detection score [4, 5]. Adopting that interface, this work examines what crop-row guidance accuracy is achievable with a pipeline that has no trained detector and no learned weights of any kind. The guidance estimator is assembled entirely from well-understood stock parts. An Excess-Green index [6, 7] feeds a column-peak central-row tracker; a robust quadratic fit $x = P(y)$ by Huber IRLS ($\delta = 1.345$, scale $1.4826 \cdot \mathrm{MAD}$) models the guidance line [8, 9]; the heading θ is taken as the look-ahead tangent to P and the cross-track offset e at the vehicle reference row; and a temporal stage of a sliding-window median, an exponential moving average, and a physical max-jump gate smooths both signals [10]. All components are established. The study offers an in-field validation of what this stock composition does on a real cabbage robot, rather than a new method.

The study contributes two results and one testable hypothesis within a deliberately narrow scope. The first result is a dataset: the first real forward-view cabbage-field robot guidance dataset, comprising 1482 forward-view cabbage frames at 640×480, screened from two field traverses (~20 and ~14 minutes), spanning a wide range of vegetation density at a forward-view camera pose representative of low-cost in-row navigation. The second result concerns leakage-free validation. Because the estimator has no trained component, the entire dataset serves as a pure test set with nothing for it to leak into, and the validation is structurally leakage-free in a way that detector-based pipelines, with their expensive and leak-prone agricultural train/test splits, are not. A guidance line is emitted on 100% of frames (1482/1482), and this rate is uniform across vegetation density, with per-tertile (low/mid/high greenness) values of 491/491, 487/487, and 504/504, each 100%. This figure is a *self-consistency* statement rather than an accuracy guarantee: "a row was found" means a line was *emitted*, not that it is the *correct* line. The accuracy-bearing quantity that completes this result, heading and cross-track error against a human-labelled ground truth, is the subject of ongoing annotation and is reported as [TBD pending GT labelling]. The testable hypothesis identifies the origin of robustness: the temporal stage, rather than the perception front end, is the dominant contributor to heading stability, and the labelled evaluation is built to test it. A “raw vs. fused” jump count cannot settle the question, because the max-jump gate rejects large frame-to-frame heading changes as a matter of design, which makes such counts an arithmetic consequence of the gate; they appear only as explicitly-labelled descriptive context. The hypothesis is confirmed or refuted by whether the temporal stage lowers heading *error against ground truth* on the hard, low-visibility frames, and at what false-reject cost. Reporting that result awaits the annotation, so it too stands as [TBD pending GT labelling].

The scope is one robot, one crop (cabbage), one camera pose, open-loop at 1 fps, and the claims extend no further. The measured outcome is a 100% guidance-line emission rate, uniform across vegetation tertiles. Every accuracy-bearing quantity remains pending: heading and cross-track error against human ground truth, inter-annotator agreement, error on the visually difficult (large raw-jump) frames, the false-reject rate, and the baseline (RANSAC and equal-weight least squares [22]) and temporal-ablation comparisons. Cross-track offset e is reported in pixels, as a percentage of half the image width, and in centimetres via an image-to-ground homography built from the robot’s measured camera height and tilt. The dataset and code will be made publicly available upon publication.

6.1. Future work

Three directions follow directly from the boundaries above. A full on-robot camera calibration comes first. The centimetre cross-track reported here uses an image-to-ground homography built from the robot’s measured camera height and tilt, with intrinsics reused from the sibling camera and a flat-ground assumption, so a checkerboard calibration performed on the cabbage robot itself, capturing its own intrinsics and a ground-plane fit, would remove the residual planar-ground and reused-intrinsics assumptions. A closed-loop field run is the second direction. The present study is open-loop, so it cannot capture how perception latency, controller dynamics, and terrain interact with the guidance estimate and the max-jump gate's decisions [25]. Replacing the recorded 1 fps evaluation with an on-robot trial, with wheel odometry and a steering controller tracking θ and e in real time, is the natural way to turn the structural robustness claim into a measured intervention rate. Multi-crop generalisation completes the list. The estimator is detector-free and training-free, so the identical code runs unchanged across crops and growth stages, yet validation so far covers only cabbage. Extending the dataset to additional row crops and conditions would test whether the temporal stage remains the primary source of robustness under genuine distribution shift, and would turn the single-crop validation into a systematic characterisation of when this stock pipeline can, and cannot, be relied upon.

Data and Code Availability

The forward-view cabbage-robot guidance dataset (two field runs, 1482 forward-view cabbage frames at 640×480 screened from 1 fps sampling, with the per-frame guidance logs) and the detector-free guidance code will be made publicly available upon publication. The hand-labelled ground-truth subset and the accuracy and ablation results of Table 4 will be released together once the annotation now in progress is complete. A repository link and archival DOI will be provided in the camera-ready version.

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Contributions

Conceptualization, investigation, and formal analysis: Manh V. Hoang, Nam Do and Thang M. Pham; methodology, validation: Manh V. Hoang and Thang M. Pham; software and writing: Manh V. Hoang. All authors were involved in its revision. All authors read and approved the final manuscript.

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